

Exercise 07^a: Check the idempotence of the 'closing-opening'

To verify the idempotence of the closing-opening operator, we used the image `cam_74.pgm` with a size of 1. We executed the program to use the closing-opening operator, and this generated this image:



Then we used the same closing-opening operator to generate the second image:



Afterwards, we used the code from `exercise_02b_compare.py` to compare both images. The comparison result was 1, proving that both images are identical. Therefore, we can confirm the idempotence property of the closing-opening operator, meaning that once an transformation is applied, if we use the same transformation once again, the image will not be modified:

$$CO(CO(I)) = CO(I)$$

So, the results obtained validate that the closing-opening operator behaves as the expected properties of mathematical morphology.

Exercise 07b: Idem for the 'opening-closing' alternated filters

To verify the idempotence property of the opening-closing operator, we used the image cam_74.pgm with a size of 1. We executed the program with the opening-closing that generated this image:



Then we used the same opening-closing to generate the second image:



Afterwards, we used the code from `exercise_02b_compare.py` to compare both images. The comparison result was 1, proving that both images are identical. Therefore, we can confirm the idempotence property of the opening-closing operator, meaning that once an opening-closing operator is applied, if we use the closing operator once again, the image will not be modified:

$$OC(OC(I)) = OC(I)$$

So, the results obtained validate that the opening-closing operator behaves as the expected properties of mathematical morphology.